

Course Code	: 2304CS304	Date	: 10-05-2025
Course Name	: Software Engineering	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Define software engineering and list its key characteristics. **4**
- (B)** Explain the importance of the software development life cycle (SDLC). **3**

OR

Describe the layered technology approach in software engineering.

- (C)** Explain Waterfall process model with proper diagram. **7**

OR

Explain RAD Model with proper diagram.

- Q.2 (A)** Sketch a Use Case diagram for a vending machine. **4**
- (B)** Explain the importance of a Software Requirement Specification (SRS) document. **3**

OR

Describe functional and non-functional requirements of a software projects.

- (C)** Sketch a state diagram for a telephone call. **7**

OR

Sketch a sequence diagram for book return in Library Management System.

- Q.3 (A)** Define cohesion and coupling in software design. **4**
- (B)** Explain the role of architecture design in software development. **3**

OR

Describe the importance of user interface design principles.

- (C)** Explain golden rules of User Interface Design (UID). **7**

OR

Explain Functionality, Usability, Reliability, Performance, and Supportability software quality attributes in detail.

- Q.4** **(A)** Explain the coding standards and their significance in software development. **4**
- (B)** Explain the difference between white-box and black-box testing. **3**

OR

Describe various software testing strategies used in project development.

- (C)** Explain Cyclomatic Complexity using flow graph. **7**

OR

Discuss how code review improves software quality.

- Q.5** **(A)** List the responsibilities of a software project manager. **4**
- (B)** Explain the significance of risk management in software projects. **3**

OR

Describe the COCOMO model used for cost estimation.

- (C)** Explain risk mitigation, monitoring, and management plan. **7**

OR

Explain W5HH principle in detail.
